

Confidence-Driven Region Refinement for Small Object Detection in UAV Image

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Abstract: UAV target detection has become increasingly indispensable in the fields of public safety, traffic monitoring, and disaster relief. However, there are still some problems in practical application, such as the difficulty of small target detection and the computational power limitation of edge devices. To tackle the aforementioned challenges, a confidence-driven region adaptive YOLOv5 detection method is developed in this paper. This method detects the global low-resolution of the input image quickly and obtains the rough position and category information of the target. Through the dynamic region selection mechanism driven by confidence, the differentiated reasoning strategy is implemented for the detection regions with different confidence, and only local high-resolution fine detection is carried out for the regions with medium confidence to restore the target details. The non-maximum suppression method is used to fuse the global and local detection results to obtain the final detection output. The experimental results on the VisDrone dataset show that, compared with the baseline YOLOv5 algorithm, the average accuracy of this method is improved by 3.6%, and the detection frame rate is increased from 30 to 35. On the basis of ensuring the real-time detection performance, this method effectively improves the small target detection effect in the UAV scene, and provides a feasible technical scheme for the UAV target detection task on edge devices.

Keywords: confidence-driven, region adaptive reasoning, YOLOv5, small target detection.

1. Introduction

UAV technology has been widely used in urban monitoring, traffic management, disaster relief, and other scenes, UAVs have gradually become an important carrier of information collection with the advantages of flexibility and a broad perspective. As the core task of the UAV vision system, target detection accuracy and reasoning speed directly affect the overall operation effect of the UAV system [1]. The special angle of view of UAV aerial photography leads to the characteristics of small-scale and sparse distribution of targets in the image [2]. Due to the low proportion of target pixels and fuzzy edge details, the traditional target detection algorithm is prone to the loss of effective information in the process of feature extraction, which leads to the decline of detection accuracy [3]. At the same time, most UAV platforms are equipped with edge computing devices, which have limited computing power and storage resources, and the deployment and application of complex depth detection models are greatly constrained [4]. How to achieve high-precision real-time detection under the condition of limited computing power has become a problem to be solved in the field of UAV target detection.

Current studies have analyzed the UAV small target detection technology from three aspects: multi-scale feature fusion, image slice detection, and dynamic reasoning optimization. First, by constructing multi-scale detection branches, the capacity of detecting objects across various scales is enhanced [5]. YOLOv5 algorithm further optimizes the feature fusion structure and achieves a more effective trade-off between inference efficiency and detection performance [6]. The development of deep learning target detection provides a theoretical reference for the improvement of multiscale methods [7]. Second, in the direction of image slice detection, existing studies have effectively alleviated the issue of feature information degradation for tiny objects through dividing high-resolution aerial images into several smaller regions for independent detection [8]. This method is easy to introduce duplicate detection and significantly increases the computational overhead [9]. Third, the direction of dynamic reasoning optimization. The faster R-CNN algorithm realizes the dynamic screening of candidate boxes through the regional recommendation network [10].

The existing literature has made some progress in improving the detection accuracy, but there is still a research gap in the real-time optimization under the constraint of computational force. In this paper, the small target detection in a UAV scene is taken as the research object, and a regional adaptive reasoning method based on confidence-driven is proposed. The main contributions of this paper are as follows: A regional adaptive reasoning framework based on confidence-driven is proposed.

2. Methods

As seen in Figure 1, a confidence-driven region adaptive reasoning method is proposed in this paper. The main idea is to balance the detection accuracy and computational efficiency through a hierarchical detection strategy, which is composed of four modules: global low-resolution detection, confidence-driven region selection, local high-resolution fine detection, and result fusion. First, global low-resolution detection. The main function is to quickly screen out the suspected target area in the image and provide guidance for subsequent detection.

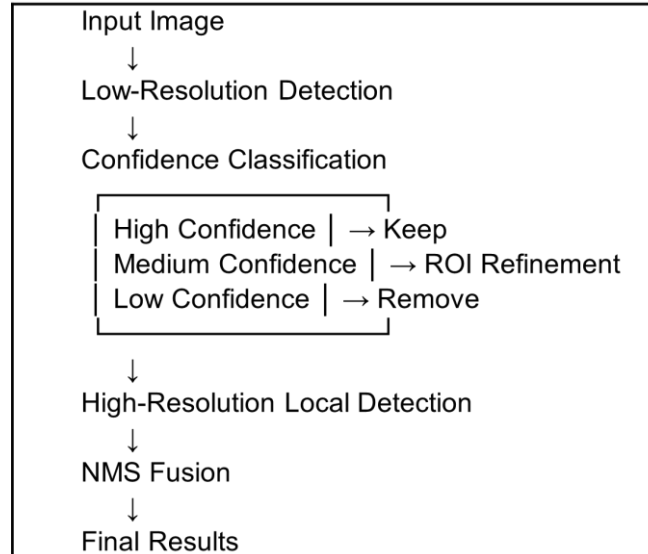


Figure 1: Regional adaptive reasoning framework (Picture credit: Original)

Second, confidence drives region selection. The subsequent processing flow of different regions is determined by grading the confidence of the global detection results. Let the detection results be defined as model (1):

$$R = \{b_i | T_l < c_i < T_h\} \quad (1)$$

Among them, b_i represents the i -th detection box, c_i represents the confidence score of the corresponding detection box, T_l and T_h represent the low confidence threshold and high confidence threshold, respectively, and R represents the set of candidate regions that need to enter the local high-resolution fine detection stage.

Third, local high-resolution fine detection. Recover the target details lost in low-resolution detection, and improve the detection accuracy of small targets. Figure 2 shows the process of local high-resolution detection.

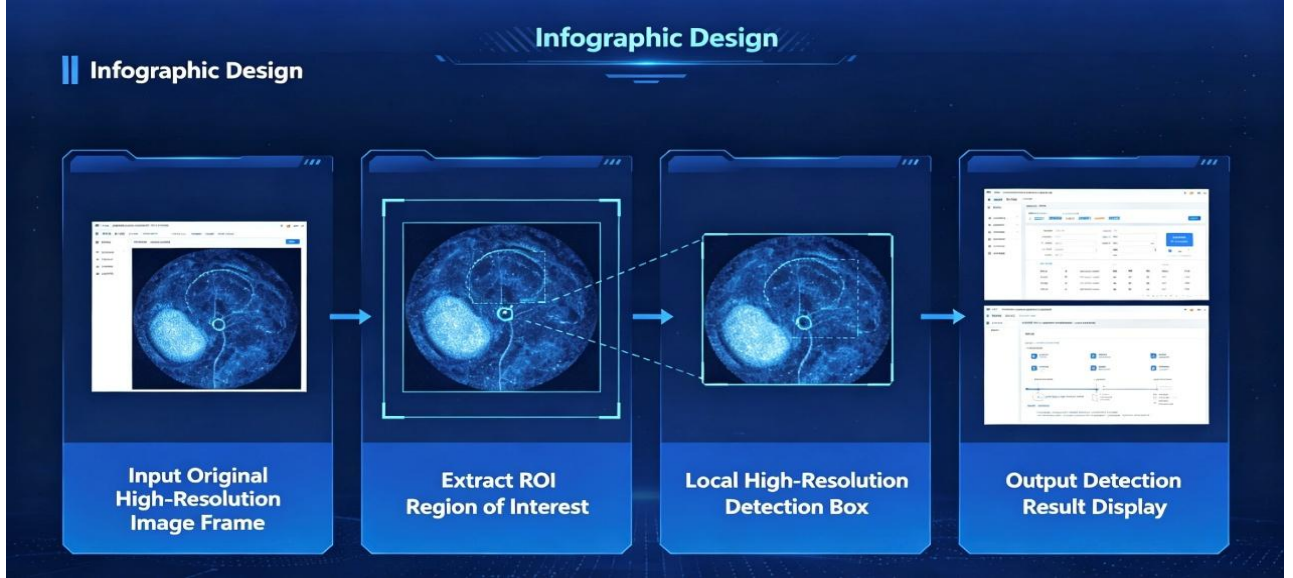


Figure 2: Local high-resolution detection process (Picture credit: Original)

Fourth, result fusion. The main task of the fusion phase is to integrate the results of global low-resolution detection and local high-resolution fine detection, eliminate duplicate detection, and get the final detection output. Since the global detection and local detection may produce multiple overlapping detection frames for the same target, the non-maximum suppression method is used for processing. The intersection over Union is defined as model (2):

$$IoU = \frac{|B_i \cap B_j|}{|B_i \cup B_j|} \quad (2)$$

Where B_i and B_j represent two candidate detection frames, respectively, and IoU represents the intersection and union ratio of the two. When the IoU between the detection boxes exceeds the set threshold, the detection box with high reliability is retained, and the duplicate box is deleted, so as to reduce the duplicate detection results.

3. Parameter settings and evaluation metrics

The experiment was verified by using the VisDrone dataset. The experiment uses the VisDrone2019-det data set, in which the training set contains 6471 images for model parameter learning and optimization. The verification set contains 548 images, which are used for model tuning and performance verification. The test set contains 1610 images, which are used to objectively evaluate the final detection performance of the model. The detection precision and real-time capability of the model are measured, respectively. The average precision includes two sub-indexes: $mAP_{0.5}$ and $mAP_{0.5:0.95}$. YOLOv5 is selected as the baseline method and compared with the confidence-driven region adaptive reasoning method proposed in this paper.

In the global detection phase, the image scaling size is 416×416 to restore the details of small targets. The ablation experiment mainly analyzes the contribution of low-resolution global detection,

confidence-driven region screening, and the local high-resolution fine detection module to the overall performance. The parameter sensitivity experiment is used to analyze the influence of different confidence threshold settings on the detection accuracy and reasoning speed.

4. Results

The overall detection performance comparison results of this method and the baseline YOLOv5 method are shown in Table 1. The experimental results show that the proposed method is superior to the baseline method in detection accuracy and inference speed by introducing the confidence-driven adaptive reasoning mechanism. In terms of detection accuracy, the mAP_{0.5} of this method is 3.6% higher than that of the baseline method, reaching 72.1%. mAP_{0.5:0.95} is 2.8 percentage points higher than the baseline method, reaching 41.3%, indicating that this method not only improves the target recognition ability but also improves the target positioning accuracy. In terms of reasoning speed, the detection frame rate of this method is increased from 30fps to 35fps, with an increase of about 16.7%. This is because the global low-resolution detection phase significantly reduces the amount of calculation. Although the local fine detection phase increases some computational overhead, the overall amount of calculation is still lower than the full image high-resolution detection, realizing the synchronous improvement of detection accuracy and calculation efficiency.

Table 1: Performance comparison between YOLOv5

Method	mAP _{0.5} (%)	FPS
YOLOv5	68.5	30
Method in this paper	72.1	35

To investigate the impact of each constituent module on the final detection effect, further ablation experiments are performed in this paper. The experimental results are shown in Table 2. The experimental results show that when only using low-resolution global detection, the model reasoning speed can be significantly improved, but the accuracy of small target detection has a certain decline. After introducing the confidence-driven region filtering mechanism, it can effectively reduce the repeated calculation of invalid regions and further improve the detection performance while maintaining a high reasoning speed. Finally, after adding the local high-resolution fine detection module, the small target details are restored, and the model achieves the optimal overall detection performance.

Table 2: Ablation results of different modules

Method	mAP _{0.5} (%)	FPS
YOLOv5	68.5	30
+Low-resolution global detection	69.4	38
+Confidence zone screening	70.3	37
+Local high-resolution fine detection	72.1	35

In order to visually show the detection effect of this method, Figure 3 shows the comparison of the detection results of the baseline method and this method in a single scene.



Figure 3: Small object detection results on VisDrone2019-DET.

Figure 3: Example of small target detection in a UAV Scene (Source: VisDrone dataset)

Figure 4 illustrates the detection results of the proposed method under various environments, including typical UAV application scenarios such as urban streets, field areas, and cloudy light.



Figure 4: Detection results in different environments on VisDrone2019-DET. Each group compares YOLOv5 (Baseline) with The Study (confidence-driven region-adaptive) under the same scene. Bounding boxes are drawn for detections with confidence score greater than 0.25. The proposed method consistently detects more small and distant objects with higher confidence across diverse conditions.

Figure 4: Example of small target detection in different environments (Source: VisDrone dataset)

The division of the confidence threshold directly determines the number of regions that need local fine detection, and then affects the detection accuracy and reasoning speed. In this paper, the high confidence threshold is set to 0.7, and the low confidence threshold is set to 0.3. Considering the requirements of detection accuracy and real-time, this paper selects 0.3 and 0.7 as the confidence

threshold, which can achieve a good balance between the two. The region adaptive reasoning mechanism proposed in this paper has good scalability and can be applied to other target detection frameworks in the future, such as Faster R-CNN or SSD, to improve their detection performance. In order to analyze the impact of different confidence threshold settings on detection performance, this paper further carried out parameter sensitivity experiments, and the experimental results are shown in Table 3.

Table 3: Detection performance under different confidence threshold settings

Low/high threshold	mAP 0.5 (%)	FPS
0.2/0.6	71.8	32
0.3/0.7	72.1	35
0.4/0.8	72.4	28

5. Conclusion

Aiming at the problems of low accuracy of small target detection and limited computing power of edge devices in the UAV scene, this paper proposes a regional adaptive YOLOv5 target detection method based on confidence-driven. By constructing a hierarchical reasoning framework combining global low-resolution fast detection and local high-resolution fine detection, the contradiction between small target feature loss and high computational overhead in traditional full image detection methods is solved. This method does not need to make complex modifications to the basic detection network, and can achieve performance improvement only by optimizing the reasoning process. It has a low deployment threshold and can adapt to the edge computing equipment carried by a UAV. It provides a feasible technical scheme for the real-time target detection task of UAVs in practical engineering.

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